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Class : SYIT	Batch : -
Date of Assignment : 23-09-26	Date/Time of Submission : 23-09-26

PART – B Practical Questions

BeautifulSoup

1.Extract Headings

Write a Python program using BeautifulSoup to extract and print all < h2> headings from a webpage.

```
pip install requests beautifulsoup4
```

```
Requirement already satisfied: requests in /usr/local/lib/python3.13/dist-packages (2.32.4)
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.13/dist-packages (4.13.5)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests) (3.4.9)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13/dist-packages (from requests) (3.19)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests) (2026.7.22)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.13/dist-packages (from beautifulsoup4) (2.9.2)
Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.13/dist-packages (from beautifulsoup4) (4.13.5)
```

```
import requests
from bs4 import BeautifulSoup

url = "https://www.geeksforgeeks.org/"
soup = BeautifulSoup(requests.get(url).text, "html.parser")

for h in soup.find_all("h2"):
    print(h.text.strip())
```

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2.Extract Links

Write a Python program using BeautifulSoup to extract all < a> tags and print their link text and URLs.

```
!pip install requests beautifulsoup4
```

```
Requirement already satisfied: requests in /usr/local/lib/python3.13/dist-packages (2.32.4)
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.13/dist-packages (4.13.5)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests) (3.4.9)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13/dist-packages (from requests) (3.19)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests) (2026.7.22)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.13/dist-packages (from beautifulsoup4) (2.9.2)
Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.13/dist-packages (from beautifulsoup4) (4.13.5)
```

```
import requests
from bs4 import BeautifulSoup

url = "https://www.google.com"
soup = BeautifulSoup(requests.get(url).text, "html.parser")

for a in soup.find_all("a"):
    print(a.text.strip(), "->", a.get("href"))
```

```
Gmail -> https://mail.google.com/mail/&ogbl
Images -> https://www.google.com/imgbp?hl=en&ogbl
-> https://www.google.com/intl/en/about/products
Sign in -> https://accounts.google.com/ServiceLogin?hl=en&passive=true&continue=https://www.google.com/&ec=GAZAmgQ
Advanced search -> /advanced_search?hl=en&authuser=0
Advertising -> /intl/en/ads/
Business Solutions -> /services/
About Google -> /intl/en/about.html
Privacy -> /intl/en/policies/privacy/
Terms -> /intl/en/policies/terms/
```

Seaborn

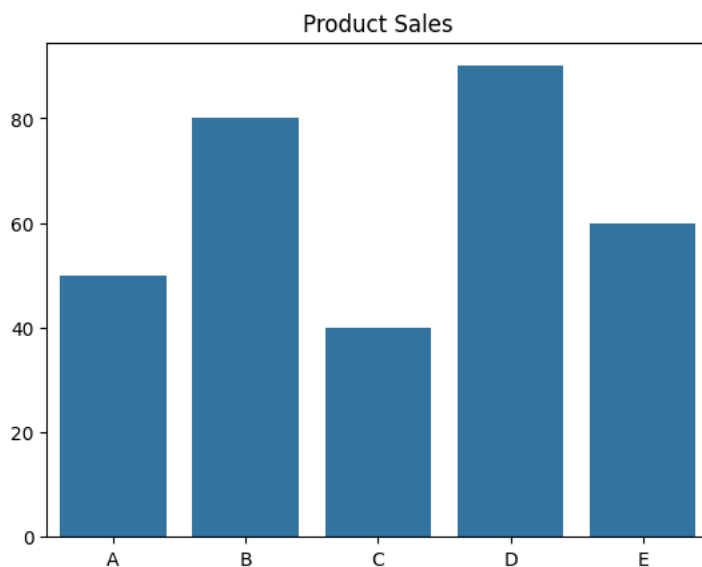
3.Bar Chart

Create a Seaborn bar chart showing the sales of 5 different products.

```
import seaborn as sns
import matplotlib.pyplot as plt

products = ["A", "B", "C", "D", "E"]
sales = [50, 80, 40, 90, 60]

sns.barplot(x=products, y=sales)
plt.title("Product Sales")
plt.show()
```



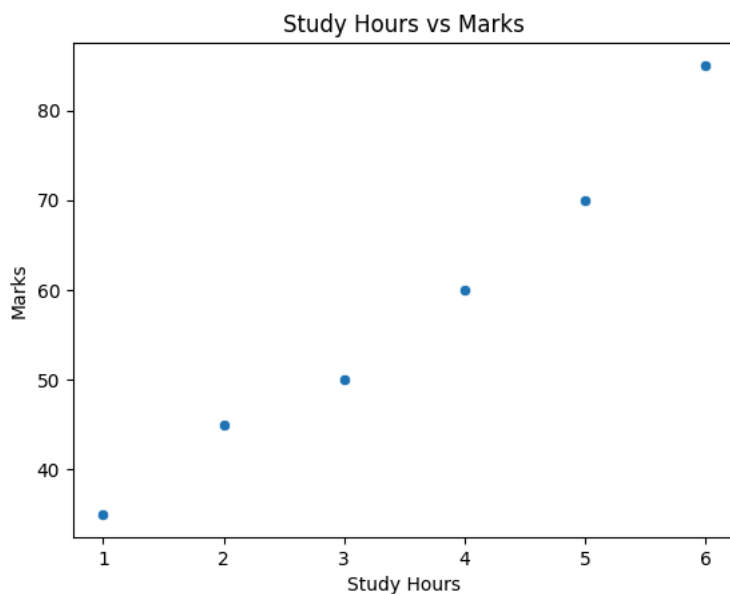
4.Scatter Plot

Create a Seaborn scatter plot showing the relationship between students' study hours

```
import seaborn as sns
import matplotlib.pyplot as plt

hours = [1, 2, 3, 4, 5, 6]
marks = [35, 45, 50, 60, 70, 85]

sns.scatterplot(x=hours, y=marks)
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Study Hours vs Marks")
plt.show()
```



5.Histogram

Create a Seaborn histogram to display the distribution of students' marks.

```
import seaborn as sns
import matplotlib.pyplot as plt

marks = [35, 45, 50, 55, 60, 65, 70, 72, 75, 80, 85, 90, 95]

sns.histplot(marks, bins=5, kde=True)
plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.title("Distribution of Marks")
plt.show()
```

